

Contrastive RL under Unobserved Confounding

Motivation: Confounding biases positive future samples

Contrastive RL uses observed transitions as learning signal:

$$(s, a) \rightarrow s^+ \sim p(s' \mid s, a) \quad r_g(s_t, a_t) \triangleq (1 - \gamma) p(s_{t+1} = s_g \mid s_t, a_t)$$

Implicit assumption: observational = causal

$$p(s' \mid s, a) \approx p(s' \mid do(a), s)$$

With confounding:

$$p(s' \mid s, a) \neq p(s' \mid do(a), s)$$

$\implies$ biased / spurious positive samples

How to apply – Causal DQN

- ▶ Partial identification: r, s'
- ▶ Solution: modify Bellman target
- ▶ Normal case: standard backup
- ▶ Low support: pessimistic backup:

$$\text{reward lower bound} + \min_{s'} Q(s')$$

- ▶ Trigger: propensity score

$$w(s, a) = \pi_b(a \mid s)$$

How to apply in Contrastive RL

- ▶ Affected object: future positive distribution
- ▶ Training: CE loss over positive / negative samples
- ▶ Pessimism: Inject pessimism into the positive term
- ▶ Support weight: propensity score

$$w(s, a)$$

Pessimistic version:

$$\begin{aligned}\mathcal{L} = & w(s, a) \cdot \log \sigma(f(s, a, s^+)) \\ & + (1 - w(s, a)) \cdot \log \sigma(f(s, a, s^{\text{worst}})) \\ & + \log(1 - \sigma(f(s, a, s^-)))\end{aligned}$$

Problems & Questions

- ▶ Alignment of problem formulation
- ▶ Direction for next step exploration

Robust version:

$$\begin{aligned}\mathcal{L}_{robust} = & - \left[w^+ \log \sigma(f(s, a, s^+)) \right. \\ & \left. + (1 - w^+) \log \sigma(f(s, a, \tilde{s}^+)) \right] \\ & - \left[w^- \log (1 - \sigma(f(s, a, s^-))) \right. \\ & \left. + (1 - w^-) \log (1 - \sigma(f(s, a, \tilde{s}^-))) \right]\end{aligned}$$